

SGAFormer: Score-Guided Attention Transformer for Gastric Section Classification

Yu-Ting Lu, Chun-Rong Huang, *Senior Member, IEEE*, Yu-Ching Tsai, and Hsiu-Chi Cheng

Abstract—Gastric section classification serves as an important preprocessing step for downstream endoscopic image analysis. However, endoscopic images of different gastric sections are visually similar, making it challenging to identify appropriate gastric sections and leading to a fine-grained classification problem. To solve the problem, we propose a score-guided attention transformer (SGAFormer), a novel architecture comprising a score-guided attention encoder backbone, a fine-grained-aware token selection module, and a pair-wise feature enhancement module. Specifically, the score-guided attention encoder backbone captures both local textures and global context while explicitly guiding the network to focus on informative regions. To distinguish subtle mucosal differences across visually similar sections and extract the most discriminative fine-grained features, the fine-grained-aware token selection module dynamically filters and aggregates multi-scale patch tokens. Furthermore, the pair-wise feature enhancement module enforces cross-image feature interaction, helping the model extract robust, diverse representations from different gastric sections. The experimental results show the superiority of the proposed method over state-of-the-art classification methods in the public and private gastric section classification datasets.

Index Terms—Transformer, Token selection, Endoscopic image, Deep learning, Gastric section classification, Computer-aided diagnosis.

I. INTRODUCTION

The upper gastrointestinal (GI) tract comprises the esophagus, stomach, and duodenum. In esophagogastroduodenoscopy (EGD), doctors use an endoscope to capture images of the stomach and duodenum and assess the severity of inflammation, malignant lesions, erosions, and ulcers. As indicated in [1]–[3], gastric sections such as the antrum, body, and cardia provide important evidence for the evaluation of histological

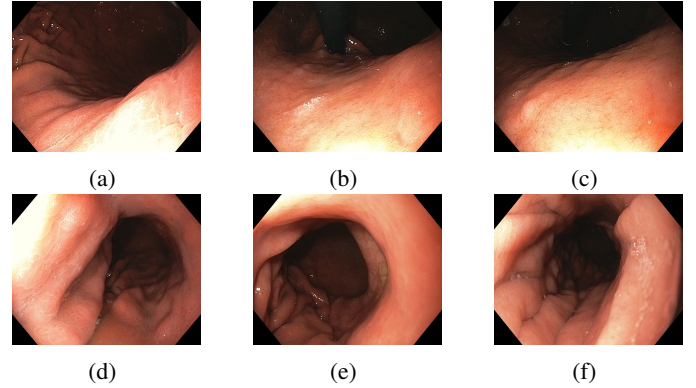


Fig. 1: Similar endoscopic images from (a) antegrade view of posterior middle upper body (P3), (b) retroflex view of lesser curvature view of fundus cardia (L4), (c) retroflex view of posterior middle upper body (P5), (d) antegrade view of anterior middle upper body (A3), (e) antegrade view of lesser curvature lower body (L2), and (f) antegrade view of greater curvature middle upper body (G3) of the upper GI tract.

parameters, including acute inflammation score, atrophy, intestinal metaplasia, and lymphoid follicles. In addition, precancerous markers such as the corpus-predominant gastritis index (CGI) [4]–[6] and gastric intestinal metaplasia (GIM) [7]–[9] also identify patients with a high risk of gastric cancer of the antrum, body, and cardia. Therefore, in EGD, determining the current location within the upper GI tract is a key aspect of the procedure.

For diagnostic purposes, the upper GI organs are divided into several sections. However, as the number of sections increases, classification difficulty also increases. Moreover, due to mucosal boundaries and changes in the field of view of the endoscopic camera, the appearances of different GI sections are visually similar, as shown in the first row and second row of Fig. 1. These visually similar endoscopic images captured from different GI sections make it difficult for the model to distinguish them. In practice, as the number of endoscopic images increases, manually selecting them for the aforementioned medical diagnosis is time-consuming. Thus, gastric section classification is an important research topic and serves as an effective pre-processing step for computer-aided diagnosis in endoscopic image analysis.

To address the challenges mentioned above, particularly the high visual similarity between different gastric sections, we formulate the gastric section classification problem as a

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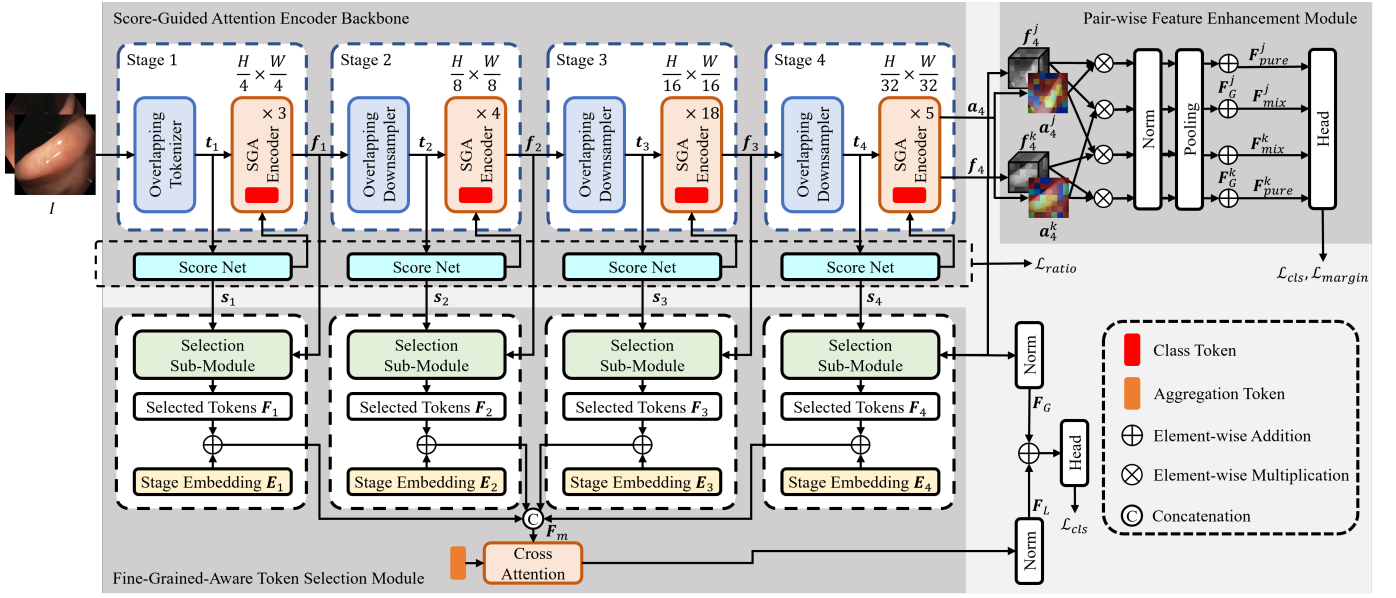


Fig. 2: The overview of score-guided attention transformer (SGAFormer), which is composed of a score-guided attention encoder backbone, a fine-grained-aware token selection module, and a pair-wise feature enhancement module.

fine-grained image classification problem. Most recent gastric section classification methods rely on simple CNN architectures, which struggle to capture the subtle discriminative features needed to distinguish visually similar endoscopic images. Existing fine-grained image classification methods are primarily designed for natural images, which focus on developing modules or techniques to extract subtle foreground features. However, endoscopic images of different gastric sections lack a clear distinction between foreground and background. Consequently, existing fine-grained methods yield limited performance when applied to gastric section classification.

To identify discriminative features between endoscopic images of different gastric sections, we propose a score-guided attention transformer (SGAFormer) as shown in Fig. 2. The proposed transformer comprises a score-guided attention encoder backbone, a fine-grained-aware token selection module, and a pair-wise feature enhancement module. The score-guided attention encoder backbone is designed to capture both local mucosal textures and global semantic context, utilizing a spatial scoring mechanism to explicitly guide the network's focus toward informative regions while suppressing irrelevant background. Building upon these features from the backbone, the fine-grained-aware token selection module selects a variety of features with diverse semantic information under different scales and aggregates them into a robust local feature. These selected features enable the model to distinguish visually similar endoscopic images. To further enhance the features extracted by the model, we propose a pair-wise feature enhancement module. By leveraging attention maps from images of different gastric sections to amplify salient features and capture more discriminative ones, the model can extract more discriminative representations.

Unlike most fine-grained image classification methods that focus exclusively on foreground features, our fine-grained-

aware token selection module comprehensively identifies semantically diverse and crucial features across multi-scale feature maps. By aggregating these selected features with global features at the backbone level, our approach effectively improves model performance. We evaluate the proposed method on the public upper GI section classification dataset [10] and a private dataset, demonstrating that it outperforms state-of-the-art methods. Our contributions are as follows:

- 1) A novel score-guided attention transformer is proposed to learn fine-grained discriminative local contextual and global semantic features.
- 2) Discriminative patch tokens are further selected by the proposed fine-grained-aware token selection module to distinguish visually similar endoscopic images.
- 3) To encourage the transformer to explore alternative discriminative cues rather than focusing solely on the most prominent patches across different gastric sections, a pair-wise feature enhancement module is proposed.
- 4) The proposed method achieves the best results among state-of-the-art methods on public and private gastric section classification datasets.

Sec. II presents the related work. The SGAFormer is described in Sec. III. Sec. IV shows the experimental results. Finally, conclusions are given in Sec. V

II. RELATED WORK

A. Gastric Section Classification

In recent years, many traditional CNN-based networks have been applied to gastric section classification. For example, Bravo et al. [11] applied ResNet [12], VGG [13], and ConvNeXt [14], Ding et al. [15] applied EfficientNet [16], and Choi et al. [17] applied SENet [18] for gastric section classification. However, traditional CNN models are not designed to solve this problem, resulting in degraded performance.

In [19], many endoscopic images of different gastric sections are found to be highly visually similar. As a result, traditional CNN models are difficult to provide representative features for classifying visually similar gastric sections.

To further improve classification performance, ensemble methods such as [20] and [21] were proposed by fusing multiple CNN networks and have shown better performance in gastric section classification. Öztürk et al. [22] proposed a long short-term memory (LSTM) architecture that can be integrated into CNNs to effectively improve accuracy. Wei et al. [23] proposed the multi-granularity collaborative classification network, which fused different fine-grained levels of features to represent endoscopic images. Nguyen et al. [24] proposed EAGLE-Net, which integrates an attention module to enhance the learned features for gastric section classification. Besides classifying gastric sections from endoscopic images, Bravo et al. [25] proposed a multi-scale sequence informative module to classify gastric sections from endoscopic videos. Nevertheless, state-of-the-art methods still fail to learn representative features to distinguish visually similar gastric sections.

B. Applications and Downstream Tasks

Accurate gastric section classification is a crucial prerequisite for various automated downstream tasks in endoscopic image analysis. For example, in the assessment of the precancerous lesion, Jhang et al. [6] utilized images of the antrum, corpus, and cardia, proposing a gastric section correlation network to model inter-regional relationships and calculate correlation losses. The detection of the other important precancerous lesion, GIM, also depends heavily on localized images of different gastric sections [9], [26]–[28]. Chen et al. [9] introduced a mask focal modulation network that guides the model to focus on critical regions, thus improving GIM segmentation results. Furthermore, Ruano et al. [29] evaluated gastric distensibility by specifically analyzing the corpus, antrum, and pylorus to capture subtle gastric wall curvatures from routine endoscopic videos. For multi-label gastric atrophy diagnosis, Cui et al. [30] explicitly divided the stomach into six regions and applied a multilevel causality learning approach to eliminate confounders and extract robust class-aware features.

Extensive research has also been dedicated to the detection of early gastric cancer [31]–[34], general gastric diseases [35]–[37], and various lesions [34], [38], [39]. For example, Liu et al. [34] designed a lightweight attention-based model for narrow band imaging (NBI), achieving accurate early detection of gastric cancer and intraepithelial neoplasia. Liu et al. [32] utilized joint diagonalization principal component analysis to boost early cancer detection efficiency, while Ding et al. [33] proposed a multimodal semantic fusion system demonstrating high sensitivity and accuracy in gastric cancer classification. Beyond disease diagnosis, region-specific gastric section classification significantly benefits functional endoscopic applications. Kim et al. [40] introduced a content-based image retrieval framework that efficiently searches for similar images based on anatomical features and contextual similarity, drastically reducing the time clinicians spend locating specific

landmark photographs. Additionally, because the limited field of view often causes biopsy sites to move out of the camera's view during endoscope navigation, Ye et al. [41] proposed an online detection cascade using a random forest classifier to achieve robust anatomical relocalization. These downstream tasks reveal the importance of gastric section classification from endoscopic images.

III. METHOD

A. Overview

As shown in Fig. 2, the model architecture is divided into a score-guided attention encoder backbone, a fine-grained-aware token selection module, and a pair-wise feature enhancement module. Given an endoscopic image I with class label y , we resize the image to $224 \times 224 \times 3$ for the score-guided attention encoder backbone. The backbone contains four score-guided attention encoder stages that extract multi-scale patch tokens f_1, f_2, f_3 , and f_4 in each stage and attention map a_4 in the fourth stage. After normalizing f_4 , we obtain the global feature F_G . For each encoder in the i th stage, a score net is used to assess the score map s_i of patch tokens and guide the encoder training. Then, the patch tokens f_i and the score map s_i serve as the input of the i th scale fine-grained-aware token selection sub-module to select representative patch tokens F_i based on global and local patch context correlations. Furthermore, to ensure that the score net accurately reflects the importance of each region, a ratio loss $\mathcal{L}_{ratio}$ is computed based on s_i .

The local patch tokens selected from different scales are concatenated to form F_m . We compute cross-attention between an additional learnable aggregation token, T_{agg} , and F_m to integrate diverse semantic information from different scales. After normalizing the output of the cross attention block, we obtain the local feature F_L . We fuse F_G and F_L to serve as the input of the classification head to compute the classification loss.

To learn the differences between endoscopic images of different gastric sections, the pair-wise feature enhancement module leverages global feature attention maps for these images. Given a pair of images I^j and I^k , we perform feature interaction using the fourth stage patch tokens f_4^j and f_4^k and fourth attention maps a_4^j and a_4^k of I^j and I^k . With the feature maps and attention maps of I^j and I^k , the pair-wise feature enhancement module generates contrastive features $F_{pure}^j, F_{mix}^j, F_{pure}^k$, and F_{mix}^k for distinguishing I^j and I^k based on the classification loss $\mathcal{L}_{cls}$ and margin ranking loss $\mathcal{L}_{margin}$. The details of these components will be introduced in the following.

B. Score-Guided Attention Encoder Backbone

The encoder backbone is modified from the neighborhood attention transformer [42], which has been shown to be more efficient than full attention-based transformers such as ViT [43]. As shown in Fig. 2, the entire backbone contains four stages with different scales. For the i th stage, an overlapping tokenizer first partitions the input into a sequence of patch tokens. Subsequently, an overlapping downsampler is applied for spatial downsampling to obtain patch tokens t_i in the i th

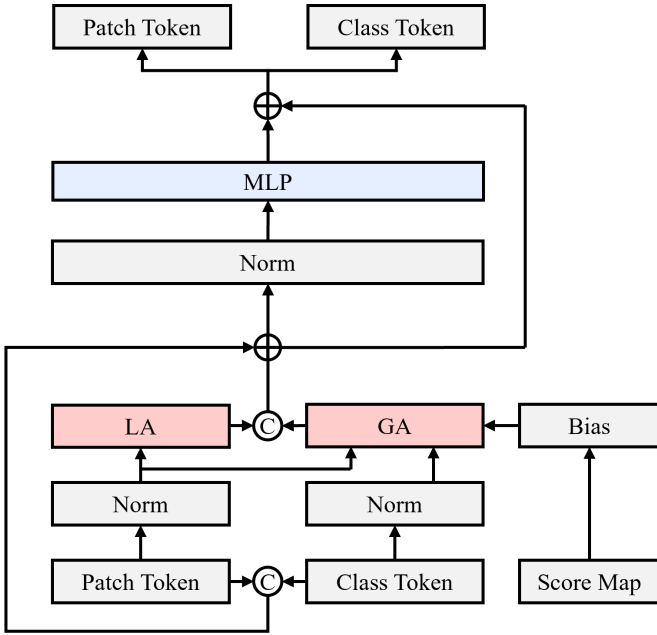


Fig. 3: The architecture of the proposed SGA encoder.

stage, and the number of channels is doubled in each stage. Although the neighborhood attention (NA) mechanism extracts features at low computational cost by restricting attention to adjacent 7×7 patches, it inherently suffers from two main disadvantages. First, its limited receptive field restricts the model from capturing long-range dependencies and global semantic context, which are crucial for understanding visually similar endoscopic images. Second, NA treats all patch tokens within the local window uniformly, lacking the explicit ability to suppress irrelevant background noise or emphasize subtle discriminative lesions.

To address these limitations, we propose a score-guided attention (SGA) encoder that leverages global context to compensate for missing global context and enables the network to focus on crucial patch tokens. To identify important patch tokens and filter out irrelevant regions, we introduce a score net in each stage i . In the score net, t_i is passed through a multi-layer perceptron (MLP) layer to extract an intermediate feature map. The map is evenly divided into two features, h_{local} and h_{global} , along the channel dimension. h_{local} is retained as local patch features, while h_{global} undergoes global average pooling across spatial dimensions to capture a global representation. Then, to explicitly capture the global and local patch context correlations, h_{local} and h_{global} are concatenated into the second MLP layer and passed through a sigmoid activation function $\sigma(\cdot)$ to predict the score map s_i as follows:

$$s_i = \sigma(\text{MLP}(\text{Concat}(h_{local}, \text{AvgPool}(h_{global})))), \quad (1)$$

where $\text{Concat}(\cdot)$ denotes the concatenation operation and $\text{AvgPool}(\cdot)$ denotes the global average pooling operation. The values of the score map s_i explicitly quantify the semantic importance of each corresponding patch token. During the subsequent attention computation, this score map serves as a crucial spatial guidance signal. Specifically, it explicitly guides the class token to recognize informative patch tokens, assign-

ing them higher attention weights while actively disregarding unimportant patches.

The proposed SGA encoder includes global attention (GA) and local attention (LA) blocks, as shown in Fig. 3. The global attention block introduces a learnable class token that interacts with the patch tokens through a cross-attention mechanism to capture the global semantic context. In this global attention computation, the class token serves as the query, while the patch tokens act as the key and value. To effectively suppress noise, we leverage the score map s_i to compute a spatial attention bias b_i as follows:

$$b_i = \log(s_i + \epsilon), \quad (2)$$

where $\log(\cdot)$ denotes the element-wise natural logarithm and ϵ is a small constant ensuring numerical stability. By directly integrating b_i into the cross-attention mechanism, we can precisely guide the model's focus toward subtle discriminative regions while preserving the original feature distribution. Consequently, the enhanced global attention is formulated as follows:

$$\text{GA}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}} + b_i\right)V, \quad (3)$$

where Q , K , and V denote the query, key, and value matrices, respectively, d represents the channel dimension of each attention head, and $\text{Softmax}(\cdot)$ is the softmax function.

Alongside global attention, patch tokens are also processed by local attention, which utilizes neighborhood attention [42] to learn local textures. The features after the GA and LA blocks are concatenated and added by a residual connection from the concatenation of the input patch token and class token. Subsequently, the combined feature is passed through a feed-forward network consisting of layer normalization and an MLP, followed by a second residual connection to further enhance the feature representations. Finally, the feature after the feed-forward network is split into a separate class token and patch tokens for the subsequent encoder blocks. Through this modified backbone, representative multi-scale patch tokens f_i are effectively generated at each stage i . Finally, after normalizing the output of the last stage f_4 , we obtain the global feature F_G .

C. Fine-Grained-Aware Token Selection Module

The fine-grained-aware token selection module aims to extract the local feature F_L , which provides representative features of each scale. In the module, discriminative fine-grained patch tokens are selected at each stage using the scale fine-grained-aware token selection sub-module. In the sub-module, the channel attention [44] of f_i is first extracted. Then, the sub-module subsequently selects the most informative patch tokens based on s_i and the channel attention. Because higher values in s_i represent important patch tokens, we keep the top n patch tokens to filter out irrelevant regions. To represent the source stage of the selected patch tokens in the i th stage, we incorporate a stage embedding $E_i \in \mathbb{R}^{1 \times C}$. Thus, the final output F_i of the sub-module in stage i is formulated as follows:

$$F_i = \text{Gather}(\text{Attn}(f_i), \text{Top-n}(s_i; n_i)) + E_i, \quad (4)$$

where $\text{Gather}(\cdot)$ is the function to retrieve selected patches based on the indices provided by $\text{Top-n}(\cdot)$, which returns the indices of the top n_i patch tokens based on s_i and $\text{Attn}(\cdot)$ is the channel attention function.

Following the operations of the sub-modules, the fine-grained-aware token selection module concatenates the selected patch tokens F_i from all stages to form a comprehensive multi-scale token sequence F_m :

$$F_m = \text{Concat}(F_1, F_2, F_3, F_4). \quad (5)$$

To aggregate features of different scales into a more discriminative representation, we perform cross-attention for F_m and a learnable aggregation token $T_{agg} \in \mathbb{R}^{1 \times C}$. Since the selected patch tokens from shallow stages capture fine-grained local textures and those from deep stages provide high-level semantic contexts, T_{agg} is explicitly introduced to adaptively fuse these hierarchical cues. Consequently, it learns a comprehensive representation that encapsulates all essential discriminative information of the endoscopic image. Here, T_{agg} acts as the query, while the F_m serves as the key and value as follows:

$$\text{CrossAttn}(Q_{agg}, K_m, V_m) = \text{Softmax}\left(\frac{Q_{agg}K_m^T}{\sqrt{d}}\right)V_m, \quad (6)$$

where Q_{agg} , K_m , and V_m denote the query, key, and value matrices, respectively, and d is the head dimension. The output of this cross-attention computation is subsequently passed through a normalization layer to produce the local feature F_L . Finally, to form the classification representation, the global feature F_G obtained from the backbone and the local feature F_L are fused via a weighted summation as follows:

$$F_{fuse} = \alpha_G F_G + \alpha_L F_L, \quad (7)$$

where α_G and α_L are hyperparameters to balance the contributions of the global and local features and are evaluated in Sec. IV-D.

D. Pair-Wise Feature Enhancement Module

To better understand the differences between endoscopic images of different gastric sections, a pair-wise feature enhancement module is proposed. To enhance feature diversity and strengthen the feature extraction capability of the backbone, we facilitate mutual interaction between patch tokens and attention maps of endoscopic images I^j and I^k of different gastric sections. The primary motivation is to utilize attention maps from different endoscopic images to strategically amplify specific patch tokens. Two different enhanced features are proposed. The first one is a pure feature, composed of the patch token and the attention map of the same endoscopic image, to enhance intra-image representation. For visually similar images of different gastric sections, projecting the spatial attention of one image onto another yields atypically enhanced features. The second one is a mixed feature, composed of the patch token and the attention map of the endoscopic images from different gastric sections, to enhance inter-image discriminability. By contrasting pure and mixed features, the model is compelled to identify subtle, fine-grained discriminative cues

and recognize the distinct characteristics of these specifically amplified tokens.

Specifically, the module constructs four distinct pure and mixed features F_{pure}^j , F_{mix}^j , F_{pure}^k , and F_{mix}^k from patch tokens and attention maps from the fourth stage of I^j and I^k . To obtain F_{pure}^j of I^j , f_4^j is multiplied element-wise by its corresponding a_4^j to further highlight salient self-tokens because F_{pure}^j aims to enhance the representation in the image of I^j . The multiplication is processed by normalization and global average pooling layers and added by the corresponding global feature F_G^j via a residual connection as follows:

$$F_{pure}^j = \text{AvgPool}(\text{Norm}(f_4^j \odot a_4^j)) + F_G^j, \quad (8)$$

where $\odot$ denotes the element-wise multiplication, $\text{AvgPool}(\cdot)$ is the global average pooling operation, and $\text{Norm}(\cdot)$ represents the normalization operation. Similarly, F_{pure}^k of I^k is obtained in the same way but multiplied by a_4^k and added by F_G^k as follows:

$$F_{pure}^k = \text{AvgPool}(\text{Norm}(f_4^k \odot a_4^k)) + F_G^k. \quad (9)$$

Unlike pure features, mixed features combine a patch token and the attention map of different endoscopic images from different gastric sections. Thus, the mixed feature F_{mix}^j of I^j is defined as follows:

$$F_{mix}^j = \text{AvgPool}(\text{Norm}(f_4^j \odot a_4^k)) + F_G^j. \quad (10)$$

and the mixed feature F_{mix}^k of I^k is defined as follows:

$$F_{mix}^k = \text{AvgPool}(\text{Norm}(f_4^k \odot a_4^j)) + F_G^k. \quad (11)$$

This cross-image operation not only yields more diverse features but also increases classification difficulty when using an attention map from a different image, effectively forcing the backbone network to enhance its feature extraction capabilities. During training, these four features will be used to compute the classification loss and margin ranking loss, which guide the model to distinguish endoscopic images of different gastric sections.

E. Loss Function

To effectively train the proposed model, we use three loss functions: the classification loss $\mathcal{L}_{cls}$, the margin ranking loss $\mathcal{L}_{margin}$, and the ratio loss $\mathcal{L}_{ratio}$. For an input image I^j , three features are generated, including F_{fuse}^j , F_{pure}^j , and F_{mix}^j . These features are fed into the shared-weight classification head to obtain their corresponding predictions, $\hat{y}_{fuse}^j$, $\hat{y}_{pure}^j$, and $\hat{y}_{mix}^j$, respectively. The classification loss $\mathcal{L}_{cls}$ is formulated as follows:

$$\mathcal{L}_{cls} = \text{CE}(\hat{y}_{fuse}^j, y) + \text{CE}(\hat{y}_{pure}^j, y) + \text{CE}(\hat{y}_{mix}^j, y), \quad (12)$$

where $\text{CE}(\cdot, \cdot)$ represents the cross-entropy function. Here, $\text{CE}(\hat{y}_{fuse}^j, y)$ serves as the guidance for the main classification task. Meanwhile, the terms $\text{CE}(\hat{y}_{pure}^j, y)$ and $\text{CE}(\hat{y}_{mix}^j, y)$ ensure that the model can still predict the correct class even when the features are modulated by an attention map of a different class, thus improving the robustness.

Because endoscopic images from different gastric sections can be visually similar, their attention maps are also similar.

TABLE I: Comparisons with state-of-the-art methods.

Method	Type	GastroHUN				AIGNS			
		Acc	F1-Score	Recall	Precision	Acc	F1-Score	Recall	Precision
ResNet152 [12]	CNN-based	85.06	84.84	84.62	85.87	90.81	87.31	86.78	88.05
ConvNext-B [14]		87.42	86.92	87.26	87.64	92.04	88.89	88.92	89.04
Swin-B [45]	Transformer-based	87.42	86.90	87.62	86.80	92.14	88.50	88.77	88.43
MViTv2 [46]		87.79	87.24	87.19	88.17	92.09	88.92	88.86	89.23
NAT [42]		88.29	87.96	88.02	88.74	92.24	89.01	88.86	89.29
TransFG [47]	Fine-grained-based	87.17	86.47	86.37	87.01	91.26	87.75	87.44	88.46
IELT [48]		86.30	85.84	85.98	86.18	91.23	87.77	87.80	87.87
MPSA [49]		87.42	86.82	86.70	87.54	92.29	89.44	89.24	89.73
Proposed	Fine-grained+Transformer	91.41	91.08	90.95	91.73	94.03	91.48	91.38	91.64

In such cases, the distinction between the pure features (e.g., F_{pure}^j and F_{pure}^k) and the mixed features (e.g., F_{mix}^j and F_{mix}^k) might be marginal. To address this issue, we introduce the margin ranking loss $\mathcal{L}_{margin}$ to explicitly distinguish the predictions of pure and mixed features. Let p_{pure}^j and p_{mix}^j denote the predicted probabilities of the ground-truth class for I^j derived from F_{pure}^j and F_{mix}^j , respectively. Similarly, p_{pure}^k and p_{mix}^k represent the corresponding probabilities of the ground-truth class for I^k . We force the discrepancy between the pure and mixed predicted probabilities to be strictly greater than a predefined margin β as follows:

$$\mathcal{L}_{margin} = \max(0, \beta - (p_{pure}^j - p_{mix}^j)) + \max(0, \beta - (p_{pure}^k - p_{mix}^k)), \quad (13)$$

where β is empirically set to 0.05. This bi-directional margin constraint effectively encourages the model to discern subtle inter-class differences. By ensuring that the network maintains significantly higher confidence in pure representations than in mixed ones, it forces the backbone to actively separate the features of visually similar images of different gastric sections.

To explicitly control the proportion of informative regions and regularize the activation magnitudes, we propose a ratio loss $\mathcal{L}_{ratio}$ based on score maps at every stage. Instead of only penalizing over-activation, we compel the average score map value at each stage to approximate a predefined target ratio. Let N_i be the total number of patch tokens in the i th stage. $\mathcal{L}_{ratio}$ is formulated as the mean squared error between the mean of the score map and the target ratio t as follows:

$$\mathcal{L}_{ratio} = \frac{1}{4} \sum_{i=1}^4 \left(\frac{1}{N_i} \sum_{m=1}^{N_i} s_{i,m} - t \right)^2, \quad (14)$$

where $s_{i,m}$ represents the value of the m th patch in the score map s_i , and $t = 0.3$. By computing the squared difference, this loss ensures that the average activation level of s_i closely matches t , effectively maintaining an optimal balance between feature sparsity and distinctiveness across different network depths.

Finally, the overall loss function $\mathcal{L}$ for training the network is the weighted sum of the aforementioned losses as follows:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{cls} + \lambda_2 \mathcal{L}_{margin} + \lambda_3 \mathcal{L}_{ratio}, \quad (15)$$

where $\lambda_1 = 2$, $\lambda_2 = 1$, and $\lambda_3 = 10$ are hyperparameters that balance the contributions of the respective loss terms.

IV. EXPERIMENTAL RESULTS

A. Datasets

We evaluated our method on the public gastric section classification dataset GastroHUN [10] and private gastric section classification dataset AIGNS. The GastroHUN dataset [10] divides the upper GI tract into 22 classes and one “NA” class. It includes 4515 training images and 803 testing images. The ground-truth labels for this public dataset were independently annotated by two professional teams, and the final labels were determined through rigorous consensus.

The AIGNS dataset collected by National Cheng Kung University Hospital under IRB B-ER-113-392, which provides informed consent to patients, divides the upper GI tract into 8 sections, including 1 esophagus section, 6 gastric sections, and 1 duodenum section, and one “none” section. For the diagnosis of precancerous lesions, not only white-light images but also narrow-band images (NBI) are collected from esophageal and gastric sections. Thus, the dataset contains 16 classes and includes 15847 training images and 3970 testing images. The ground-truth labels were provided by a single professional physician. The same as [10], [23], [24], we used Top-1 accuracy (Acc), macro F1-score (F1-Score), macro recall (Recall), and macro precision (Precision) to evaluate the performance of the proposed method and competing methods.

B. Implementation Details

We resized all images to 224×224 to maintain consistency with the network’s dimensions. The method was run on a GTX 4090 GPU. We used the AdamW optimizer [50] for model training. To preserve the pre-trained knowledge in the backbone, we apply a smaller learning rate of 10^{-5} , while the newly initialized parameters are updated with a learning rate of 10^{-4} . Furthermore, the learning rates are adjusted using a cosine annealing schedule with a linear warmup. The training epochs are set to 200. To implement the pair-wise feature enhancement module, we randomly select two classes per batch and sample 10 images from each class. Overall, the four stages of our backbone consist of 3, 4, 18, and 5 SGA encoders, respectively. The number of top- n_i tokens selected at each stage is set to 256, 128, 64, and 32, respectively. Based on our experiments, we set $\alpha_G = 0.5$ and $\alpha_L = 1$, and the evaluations of these parameters are shown in Sec. IV-C.

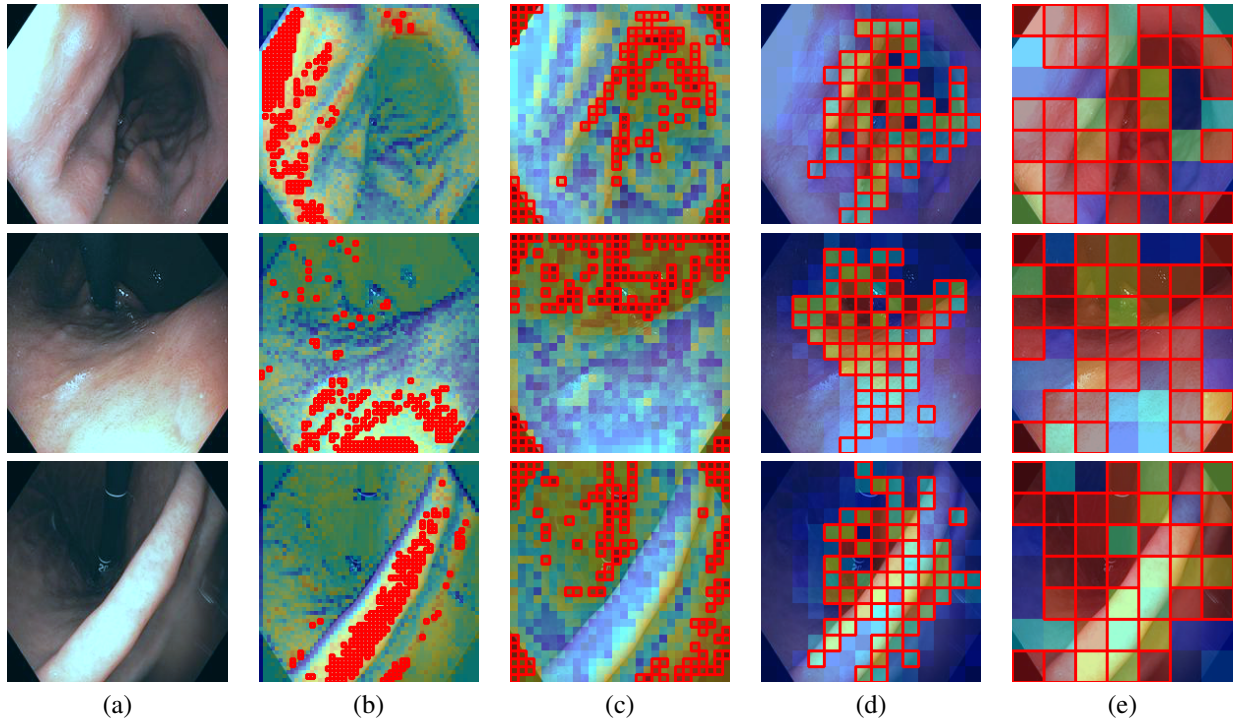


Fig. 4: Visualizations of fine-grained token selection module and score maps for visually dissimilar gastric sections. (a) Original images, (b) heatmaps for Stage 1, (c) heatmaps for Stage 2, (d) heatmaps for Stage 3, and (e) heatmaps for Stage 4.

C. Quantitative Results

Table I presents quantitative comparisons of the proposed method against various state-of-the-art models on the GastroHUN and AIGNS datasets, including accuracy, F1-Score, recall, and precision. Transformer-based models generally outperform CNN-based models. This demonstrates that the self-attention mechanism is better at capturing long-range dependencies and the global semantic context needed to understand complex endoscopic scenes. While fine-grained-based methods are designed to capture local details, they surprisingly underperform some transformer-based models in our task. This performance gap highlights a crucial domain-specific challenge. Unlike natural images (e.g., birds or cars), endoscopic images lack a clear foreground-background distinction. Consequently, focusing solely on extracting foreground local patches may cause these fine-grained methods to discard surrounding mucosal textures that actually contain vital contextual clues.

To overcome these limitations, the proposed method combines the strengths of fine-grained token selection and global context preservation simultaneously. Furthermore, it significantly enhances feature extraction capabilities through the pair-wise feature enhancement module. On the GastroHUN dataset, our method achieves an accuracy of 91.41%, surpassing the second-best method 88.29% by a significant margin of 3.12%. Furthermore, our method achieves the highest scores across all metrics, indicating its robustness in classifying visually similar gastric sections. Finally, to demonstrate the generalization capability of our architecture, we present the results on the AIGNS dataset. Consistent with previous observations, the proposed method again yields the best performance, achieving a Top-1 Accuracy of 94.03% and

an F1-Score of 91.48. This consistent superiority across two datasets demonstrates that our explicitly designed modules can effectively and universally extract the most discriminative features for gastric section classification, even under different lighting conditions, i.e., white-light and NBI images.

D. Qualitative Results

To better explain the internal mechanisms of our model, we visualize the token selection process and the generated score maps s_i at each stage using the GastroHUN dataset. As shown in Fig. 4, we present heatmaps of the score maps, along with the corresponding selected tokens, across four stages for various gastric sections. The red rectangles represent the selected patch tokens using the fine-grained-aware token selection module. It can be observed that the s_1 and s_2 effectively capture the fundamental contours and edges of the anatomical structures as shown in Fig. 4(b) and (c). The difference of highly activated regions between s_1 and s_2 indicates that the fine-grained-aware token selection module effectively gathers diverse, complementary multi-scale features, preventing the network from over-relying on a single localized region for decision-making. In s_3 , the selected patch tokens concentrate on the critical regions that are essential for classification, as shown in Fig. 4(d). The selected patch tokens in s_4 demonstrate that our module can select patches with semantic information using the score map. Moreover, the attention bias integrated into our backbone helps suppress relatively unimportant background regions so that the selected patch tokens can be more representative for different gastric sections, which is particularly evident in s_3 .

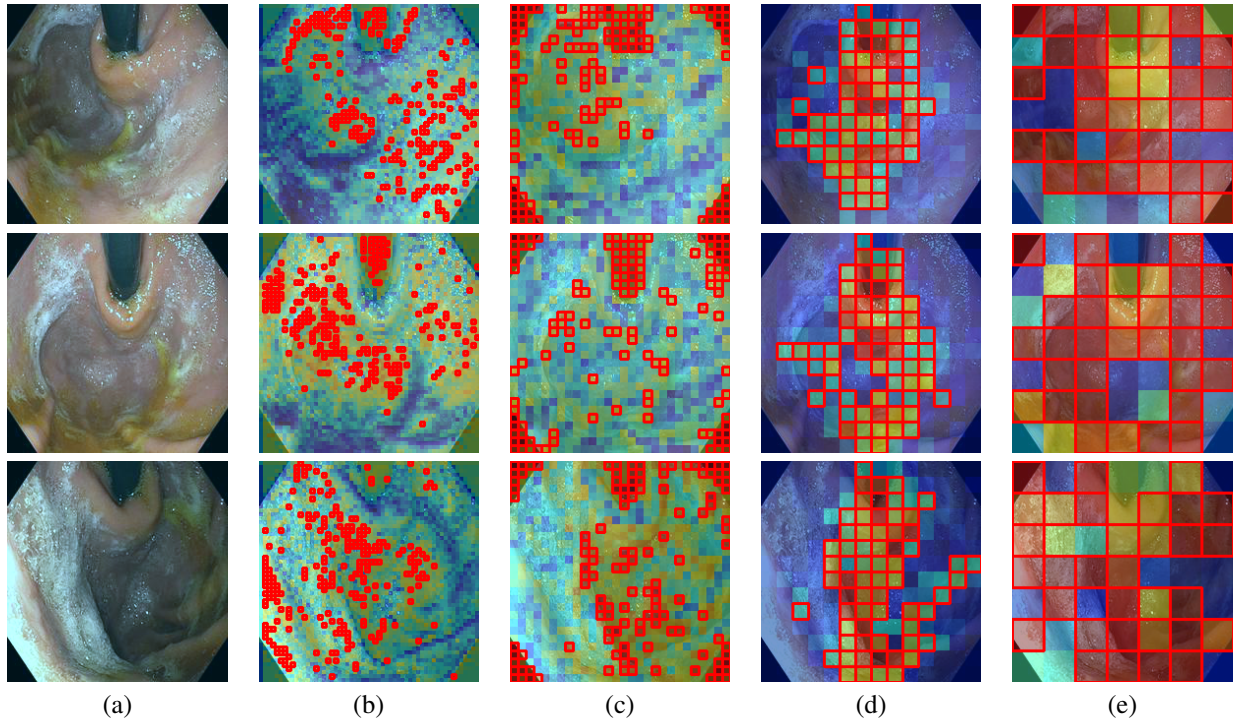


Fig. 5: Visualizations of fine-grained token selection module and score map for visually similar gastric sections. (a) Original images, (b) heatmaps for Stage 1, (c) heatmaps for Stage 2, (d) heatmaps for Stage 3, and (e) heatmaps for Stage 4.

TABLE II: Ablation study of the proposed modules.

SGA	FTSM	PFEM	T_{agg}	Accuracy (%)
✓				89.54
✓	✓		✓	90.29
✓		✓		90.16
✓	✓	✓	✓	91.41
✓	✓	✓		90.16

Fig. 5 illustrates the score map visualizations for visually similar endoscopic images belonging to different gastric sections. Specifically, the three example images all depict the fundus cardia but from different endoscopic views. From Fig. 5, our fine-grained-aware token selection module selects different patch tokens. The selected patch tokens further aggregate hierarchically across stages to distinguish these three visually similar endoscopic images under the specific viewing angles. Our pair-wise feature enhancement module facilitates cross-image interaction using the stage 4 attention maps and patch tokens. It helps discover discriminative features rather than common, irrelevant features shared across images.

E. Ablation Study

To evaluate each proposed module and the sensitivity of hyperparameters, we performed extensive ablation studies.

Effectiveness of proposed modules. We first investigate the SGA encoder backbone and the combined effects of the fine-grained-aware token selection module (FTSM) and the pair-wise feature enhancement module (PFEM). As shown in Table II, utilizing the SGA backbone alone yields an accuracy of 89.54%. Introducing the FTSM improves performance

TABLE III: Ablation study on the number of tokens per stage.

Top-n tokens per stage	Accuracy (%)
[5, 5, 5, 5]	90.16
[10, 10, 10, 10]	90.29
[20, 20, 20, 20]	89.66
[32, 32, 32, 32]	90.16
[64, 32, 16, 8]	90.66
[256, 128, 64, 32]	91.41
[313, 78, 19, 4]	90.29
[2048, 512, 128, 32]	90.66

to 90.29%, confirming that our token selection mechanism successfully identifies critical localized patches as shown in Table II. Similarly, using the PFEM without FTSM can also improve performance, indicating that cross-image feature interactions effectively enhance the backbone’s representational capacity as shown in the third row of Table II. When both FTSM and PFEM are integrated with the SGA encoder backbone, the model achieves a substantial performance leap, reaching the highest accuracy of 91.41%. We also evaluated the necessity of the learnable aggregation token (T_{agg}) in the FTSM. When removing T_{agg} , the accuracy decreased to 90.16%, proving that T_{agg} successfully acts as a crucial query to adaptively fuse hierarchical clues and extract a more comprehensive multi-scale representation.

Impact of the number of selected tokens per stage. Table III illustrates the effect of varying the number of selected tokens across the four stages. We compare uniform token selection strategies (e.g., [32, 32, 32, 32]) against hierarchical decreasing strategies. The results demonstrate that allocating

TABLE IV: Ablation study on the fusion weights α_G and α_L for F_{fuse} .

α_G	α_L				
	0.1	0.3	0.5	1	2
0.1	89.29	90.16	89.04	90.03	89.66
0.3	89.29	90.29	90.04	90.41	90.54
0.5	91.03	91.03	90.78	91.41	90.91
1	90.04	90.54	90.66	90.41	90.16
2	90.41	90.28	90.04	90.16	90.66

TABLE V: Ablation study on the loss functions.

$\mathcal{L}_{cls}$	$\mathcal{L}_{margin}$	$\mathcal{L}_{ratio}$	Accuracy (%)
✓			89.17
✓		✓	90.16
✓	✓		90.54
✓	✓	✓	91.41

more tokens to shallow layers and fewer to deep layers generally yields superior performance. This aligns with our architectural design: shallow stages possess larger spatial dimensions and capture abundant low-level edge and texture details, whereas deeper stages encode highly abstract, global semantics. The specific configuration of [256, 128, 64, 32] tokens achieves the optimal accuracy.

Analysis of fusion weights. In our model, the final classification representation F_{fuse} is formulated by combining the global feature F_G and the local feature F_L using hyperparameter weights α_G and α_L in Eq. (7). Table IV presents the grid search results for these weights. The model reaches its peak performance when $\alpha_G = 0.5$ and $\alpha_L = 1$. This weight distribution intuitively indicates that while the global context ($\alpha_G = 0.5$) provides a necessary structural foundation, the fine-grained local features ($\alpha_L = 1$) extracted by our cross-attention mechanism play a more dominant role in distinguishing complex anatomical sites.

Effectiveness of loss functions. To validate the necessity of our optimization strategy, we ablate the combination of the proposed loss functions, the classification loss $\mathcal{L}_{cls}$, the margin ranking loss $\mathcal{L}_{margin}$, and the ratio loss $\mathcal{L}_{ratio}$. As shown in Table V, optimizing the network solely with $\mathcal{L}_{cls}$ yields a baseline accuracy of 89.17%. Integrating $\mathcal{L}_{ratio}$ improves the performance to 90.16%. This indicates that explicitly supervising the score maps with the ratio constraint effectively guides the network to accurately evaluate token importance and retain the most crucial contextual information. Similarly, applying $\mathcal{L}_{margin}$ without $\mathcal{L}_{ratio}$ also brings a solid improvement, raising the accuracy to 90.54% as shown in Table V. Verify that the bi-directional margin constraint effectively forces the model to separate visually similar representations. Finally, combining all three losses achieves the optimal accuracy as shown in the last row of Table V.

V. CONCLUSION

In this paper, we propose SGAFormer to address the challenging task of gastric section classification. By introducing score-guided attention encoders, our model effectively captures global context while suppressing irrelevant backgrounds.

Furthermore, by dynamically filtering multi-scale patches with the fine-grained-aware token selection module, we successfully extract the most discriminative fine-grained tokens that capture subtle mucosal details across different gastric sections. Finally, to better understand the differences between endoscopic images of different gastric sections, the pair-wise feature enhancement module is incorporated for cross-image feature interaction. By combining refined local information from selected patch tokens with global semantic context, the proposed SGAFormer effectively distinguishes visually similar anatomical sites across varying viewpoints, outperforming state-of-the-art methods. In the future, we will explore the applications of the proposed architecture to other downstream endoscopic image analysis tasks, such as lesion detection [51] and segmentation [52] of different gastric sections.

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